

SmartHarvest - EO-driven Vineyard Monitoring System Specification Document (SSD)

1. Problem Definition

1.1 Business problem

Vineyard managers and agronomists require monitoring that is repeatable, timely, and interpretable in order to:

- detect within-parcel heterogeneity (zones evolving differently, therefore requiring prioritized scouting),
- detect actionable changes and emerging anomalies between subsequent satellite acquisitions (e.g., new stress patterns, unexpected outliers, or sudden shifts in spatial structure),
- assess maturation uniformity to support harvest decisions and targeted interventions.

Current practice relies on manual scouting and fragmented GIS workflows. As a result, it is difficult to elaborate a systematic and consistent analysis, particularly under time constraints and when EO observations are degraded by clouds or other quality issues.

1.2 Vision

SmartHarvest provides EO-first monitoring of vineyard parcels through an on-demand workflow within an intuitive environment that unveils field evolution states over time. The system produces anomaly maps and parcel-level summaries, explicitly reporting confidence to prevent over-trust in low-observation conditions and to support practical scouting and management decisions, at present left to the user.

1.3 ML formulation and learning problem statement

The MVP operates in an unsupervised setting. No labeled anomalies are required to produce outputs. Optional expert feedback may be captured to support evaluation and refinement in the future.

Inputs

EO observations and additional geospatial layers over a user-defined Area of Interest (AOI). Each run is anchored to a specific satellite acquisition.

Core tasks

- Spatial structure discovery and anomaly detection (scene-based): identify within-parcel structures (clusters/zones) and detect outliers or anomalous regions relative to the parcel's dominant spatial pattern for the target acquisition.

- Temporal evolution monitoring (tracking-based): track the persistence, emergence, merge/split, and drift of the discovered zones across successive acquisitions, enabling a time-navigation view of field evolution without requiring seasonal historical references.

Outputs

- a per-pixel anomaly layer on a 10 m grid (and parcel-level aggregates),
- a parcel-level maturation uniformity report derived from spatial structure,
- a confidence label and reliability surface derived from observation adequacy and EO quality indicators.

Ground truth strategy

The MVP does not require labeled anomalies for operation. When available, expert feedback may be stored as confirmations/rejections or qualitative notes at parcel level for evaluation and future governance, without being a dependency for run completion.

1.4 Key Performance Indicators (KPIs)

KPIs are grouped into Product, System Completeness, Robustness and Traceability and MLOPs dimensions to reflect user value and operational verifiability, and to maintain alignment with the Project Proposal success criteria.

Product KPIs

- **KPI-01 Run latency:** The time from run submission to a terminal state (SUCCESS / FAILED / LOW_CONFIDENCE_DONE) is ≤ 60 minutes in the test environment (p95).
- **KPI-02 Usability:** In a guided test session using the provided deployment, a new user completes: create account, save/import parcel, start run, view results, download exports within ≤ 30 minutes, without developer intervention.

System completeness KPIs

- **KPI-03 Output completeness:** For every run that reaches a terminal state, the run bundle contains the three mandatory outputs (heatmap, macro report, coverage-reliability factor) and the three exports (PDF, CSV, GeoJSON).
- **KPI-04 Policy compliance:** For every run, 100% of anomalies labeled as “confirmed” satisfy the ensemble confirmation rules; violations are detectable as failing test cases and must result in non-confirmed outputs.

Robustness KPIs

- **KPI-05 Low-confidence correctness:** Whenever observation adequacy is below the defined threshold, the run ends in LOW_CONFIDENCE_DONE, produces the mandatory outputs and exports, labels zero anomalies as confirmed, and the low-confidence trigger is explicit in report and UI.

Traceability and MLOps KPIs

- **KPI-06 Run reproducibility:** Given a run identifier, the system can retrieve the full configuration and dataset provenance (time window, AOI hash, data sources / scene

references when available, pipeline version) required to re-execute the run under the same conditions. When re-executed, the system produces the same output.

2. Data Specification

This section defines the MVP data inputs, the derived logical data products, and the minimum data-quality and provenance rules required to keep run outputs auditable, reproducible, and conservative under missing or degraded EO observations.

2.1 Data sources

The MVP relies on open Earth Observation time series complemented by static geospatial layers, plus a user-provided AOI boundary.

Baseline sources:

- Sentinel-2 L2A (optical surface reflectance), used to derive vegetation and moisture-related indices (NDVI/NDRE/NDMI) as the primary parcel-scale signal.
- SRTM DEM (static topography), used to derive terrain descriptors (slope/aspect and simple exposure proxies) to contextualize spatial variability.
- AOI boundary (parcel geometry) provided through UI drawing or GeoJSON import.

Other sources:

- Sentinel-1 SAR, as a robustness complement when optical data are cloud-limited.
- Landsat 8/9 thermal / LST and ECOSTRESS LST, as thermal context features when availability is sufficient.
- ERA5-Land, as coarse climatic context (aggregated to match run windows).

All mentioned inputs are open datasets, and each run records dataset provenance (dataset identifiers, acquisition time range, and scene references when available) as part of the run metadata, enabling traceability and re-execution.

2.2 Run datasets

Each run is executed as an on-demand analysis that constructs a run-scoped EO dataset by querying Google Earth Engine through the platform APIs. The run dataset is defined as a rolling, recency-based snapshot of the selected parcel: by default, it covers the most recent three months of available Earth Observation data at the time the run is triggered. This design ensures that the analysis remains aligned with operational monitoring needs and that results reflect the latest observations, while remaining fully traceable through run metadata.

1) RUN-EO dataset

For the selected AOI, the system retrieves and assembles all eligible observations within the last three months (relative to run submission time), applies the minimum preprocessing and QA rules (masking, harmonization, validity statistics), and produces a harmonized set of features ready for the ML kernel.

The dataset is multi-source by construction: depending on availability and the configuration of the run, it may integrate observations from one or more satellite sources. The dataset is therefore refreshed implicitly at each run execution.

2) RUN-FEATURE dataset

From the RUN-EO dataset, the system derives the analytical metrics consumed by the ML kernel. This product includes the per-cell grid-aligned features computed over the rolling window, together with the associated data adequacy statistics required for confidence labeling. The resulting representation is the input for anomaly scoring, summary metrics, and any conservative post-processing rules defined by the MVP.

Both products are persisted within the run bundle and referenced through the artifact registry and run metadata. This ensures that each run remains self-contained, auditable, and reproducible, while still reflecting the most recent EO observations available at execution time.

2.3 Data flow

The data flow is deterministic and run-oriented:

1. The AOI is acquired (draw/import), validated against the AOI contract and saved for run requests.
2. Then EO and static layers are retrieved for the AOI and the target acquisition (default: the latest available Sentinel-2 overpass), identified by acquisition date and/or scene identifier.
3. Preprocessing and QA transform raw observations into a valid observation set, and compute quality statistics that quantify missingness and observation adequacy for the acquisition.
4. Feature construction produces the run datasets, which are then consumed by the analytics kernel to compute anomaly outputs, spatial structure (zones/clusters), and parcel summaries. In parallel, the system computes the coverage-reliability factor and determines whether the run must be downgraded to a low-confidence/no-decision behavior.
5. Finally, the system writes the core outputs, generates exports (PDF/CSV/GeoJSON), completes the run metadata, and serves results via the API and UI.

This flow is consistent with an on-demand operating MVP: runs are initiated by explicit user requests, and the system does not require mandatory scheduled execution.

2.4 Data quality specifications

EO monitoring is naturally exposed to incomplete observations (clouds, shadows, acquisition gaps). For this reason, data quality is treated as a first-class concern and is explicitly connected to run terminal states and final artifacts.

At minimum, the system quantifies:

- **Completeness:** observation adequacy for the target acquisition, expressed as scene availability and usable coverage.
- **Validity:** the fraction of pixels retained after masking invalid observations for the target acquisition.
- **Alignment:** confirmation that all raster products are aligned to a declared reference grid.
- **Missingness:** explicit accounting of gaps and masked coverage affecting reliability, including spatial patterns of missing/low-quality pixels within the AOI.

If observation adequacy is below a configurable minimum, the run reaches a LOW_CONFIDENCE terminal label. LOW_CONFIDENCE runs still produce the three mandatory outputs and exports, but they must not present anomalies as confirmed in a way that could mislead the user. The low-confidence reason is exposed consistently across reports, UI, and exports.

This behavior supports the project proposal's robustness and trustworthiness propositions and the success criteria.

2.5 Preprocessing specification

Preprocessing ensures that features are comparable across acquisitions and that spatial interpretation is stable at parcel level.

Preprocessing steps include:

1. optical and thermal masking using QA strategies;
2. geospatial harmonization (reprojection and resampling to the declared reference grid, with consistent handling rules for continuous and categorical layers);
3. agronomic metrics computation.

A conservative gap-handling policy is enforced: the system does not build up long missing periods and does not force reconstruction of absent observations. Any interpolation or imputation must be explicitly recorded in the run metadata so the run remains auditable and reproducible.

2.6 Data provenance, traceability, and versioning (metadata)

Each run produces a minimal set of provenance and traceability fields required for reproducibility.

Minimum metadata includes:

- AOI identifier and hash plus schema version;
- selected time window and thresholds (including any reliability thresholds);
- dataset provenance (dataset/product identifiers, acquisition time range, and scene references when available);
- preprocessing and detector/policy versions (e.g., code version identifiers);
- quality statistics (acquisition counts, valid pixel ratios, missingness indicators);
- and artifact registry entries linking to the core outputs and exports.

Details are specified in Appendix B - Run Metadata Contract.

3. Requirements

3.1 Requirements Summary

User Requirements (UR)

- **UR-01:** The user shall create/select a parcel, submit an on-demand run for the latest available acquisition (and optionally for a selected past acquisition), monitor status, view results, and download exports.
- **UR-02:** The user shall view the run outputs in a map-first dashboard (anomaly layers, macro report, reliability surface/indicator), and switch between available layers without leaving the map context.
- **UR-03:** The user shall download exports in PDF, CSV, and GeoJSON formats.
- **UR-04:** The user shall be informed when results are low-confidence due to insufficient EO observation adequacy or poor data quality for the target acquisition, and the UI shall avoid presenting anomalies as confirmed in LOW_CONFIDENCE runs.
- **UR-05:** The user shall browse historical runs for the parcel via a date/timeline selector and open past results to review field evolution over time.

External Interface Requirements

User Interface (UI)

- **UI-01:** The UI shall allow parcel creation/import (GeoJSON/Shapefile) and provide immediate validation feedback against the AOI contract.
- **UI-02:** The UI shall allow run submission for the latest available acquisition (and, if enabled, a selected past acquisition) and provide status tracking until the run reaches a terminal state.
- **UI-03:** The UI shall present outputs in a map-first layout and enable download of PDF/CSV/GeoJSON exports for the currently selected run.
- **UI-04:** The UI shall provide a date/timeline selector that lists available runs for the selected parcel and allows switching the displayed map layers to any past run without reloading the application context.
- **UI-05:** The UI shall surface observation reliability consistently (reliability indicator and/or reliability map) and visually distinguish LOW_CONFIDENCE runs from normal runs in both the run list and the map view.

API Interfaces (API)

- **API-01:** The API shall accept a run request for a parcel and a target acquisition selector (default: latest available) and return a unique run_id.
- **API-02:** The API shall return run status and a structured failure reason if FAILED, and a structured low-confidence reason if LOW_CONFIDENCE.
- **API-03:** The API shall return run outputs (map layer descriptors and artifact references), reliability indicator/surface references, and core run metadata for a given run_id.

- **API-04:** The API shall list available runs for a parcel, including acquisition date/scene identifier, terminal status, reliability summary, and references needed for UI timeline navigation.

Data Requirements (DATA)

- **DATA-01:** AOI inputs shall conform to a defined GeoJSON schema and validation rules.
- **DATA-02:** Each run shall produce a run bundle with mandatory outputs and exports, anchored to a specific acquisition (acquisition date and/or scene identifier), with stable naming and self-contained references.
- **DATA-03:** Exported GeoJSON shall be syntactically valid and include required attributes, including run_id, parcel_id, acquisition date/scene identifier, confidence label (including LOW_CONFIDENCE when applicable), and the anomaly/reliability fields required by the export contract.

System Functional Requirements (FR)

- **FR-01:** The system shall authenticate users for UI/API access.
- **FR-02:** The system shall support parcel creation/import with validation.
- **FR-03:** The system shall allow saving and retrieving parcels per user across sessions.
- **FR-04:** The system shall create a run record and unique run_id on user submission.
- **FR-05:** The system shall manage run statuses: QUEUED, RUNNING, SUCCESS, FAILED, LOW_CONFIDENCE_DONE.
- **FR-06:** The system shall ingest required EO data and static layers for the AOI and the target acquisition (default: latest available), identified by acquisition date and/or scene identifier; optional minimal contextual acquisitions may be retrieved when required for stabilization/tracking, without being a user-defined time window.
- **FR-07:** The system shall apply preprocessing and QA prior to analysis.
- **FR-08:** The system shall compute core indices/features required by the detector.
- **FR-09:** The system shall compute a pixel-level anomaly score without ground-truth labels.
- **FR-10:** The system shall compute a coverage-reliability factor for every run.
- **FR-11:** The system shall confirm anomalies only when configured consistency/ensemble rules are satisfied and observation reliability is adequate; LOW_CONFIDENCE runs shall not present anomalies as confirmed.
- **FR-12:** For each terminal run, the system shall generate the following mandatory outputs: anomaly heatmap (spatial layer), macro report, reliability factor (and/or reliability surface reference).
- **FR-13:** The system shall generate PDF/CSV/GeoJSON exports for each terminal run.
- **FR-14:** The system shall store run metadata sufficient for reproducibility and auditability, including parcel reference, acquisition date/scene identifier, quality/reliability summaries, pipeline version, and artifact references inside the run bundle.
- **FR-15:** FAILED runs shall include structured error information (stage, reason).
- **FR-16:** The system shall support historical navigation by listing available runs for a parcel (ordered by acquisition date), including terminal status, reliability summary, and references required to retrieve outputs and exports for each run.
- **FR-17:** The system shall produce and persist a tracking representation across successive runs that supports zone/cluster evolution summaries for historical interpretation in the UI.

3.2 Structured Requirements

Structured record format:

Requirement ID, Title, Description, Rationale, Dependencies, Acceptance Criteria, Priority, Verification.

User Requirements (UR) - Structured

Requirement ID: UR-01

Title: End-to-end on-demand workflow (acquisition-anchored)

Description: The user shall be able to create/select a parcel, submit an on-demand run for the latest available acquisition (and optionally for a selected past acquisition), monitor execution status, view results, and download exports.

Rationale: Defines the MVP user journey in an on-demand, acquisition-anchored monitoring model.

Dependencies: UI-01, UI-02, UI-03, FR-01, FR-02, FR-03, FR-04, FR-05, FR-12, FR-13, FR-14, NFR-05

Acceptance Criteria: In a guided demo, a new user completes the workflow within ≤ 30 minutes without developer intervention.

Traceability: SC5, D4, M3

Verification: Demonstration

Requirement ID: UR-02

Title: Map-first results visualization and layer switching

Description: The user shall view run outputs in a map-first dashboard including anomaly layers, a macro report, and a reliability surface/indicator, and shall switch between available layers without leaving the map context.

Rationale: Enables scouting prioritization and trust through immediate spatial interpretation.

Dependencies: UI-03, FR-12, FR-10, API-03

Acceptance Criteria: For any terminal run, the UI displays run status and the three mandatory outputs, and allows switching between anomaly, feature (when available), and reliability layers without navigation away from the map.

Traceability: SC5, D4, M3

Verification: Demonstration

Requirement ID: UR-03

Title: Export and share

Description: The user shall download run exports in PDF, CSV, and GeoJSON formats.

Rationale: Supports operational sharing and GIS interoperability.

Dependencies: UI-03, FR-13, DATA-02, DATA-03, API-03

Acceptance Criteria: Exports are generated and downloadable for any terminal run, and GeoJSON opens in a GIS viewer and contains the required attributes (including run_id, aoi_id, acquisition_date/scene_id, status, reliability_factor, confidence_label, and macro summary fields).

Traceability: SC2, D5, M4; SC5, D4, M3

Verification: Demonstration, Inspection

Requirement ID: UR-04

Title: Reliability awareness and low-confidence behavior

Description: The user shall be informed when results are low-confidence due to insufficient EO observation adequacy or poor data quality for the target acquisition, and the UI shall avoid presenting anomalies as confirmed in LOW_CONFIDENCE_DONE runs.

Rationale: Prevents misleading decisions under cloud coverage and degraded observation conditions.

Dependencies: UI-05, FR-10, FR-11, FR-12, NFR-04, API-02, API-03

Acceptance Criteria: LOW_CONFIDENCE_DONE runs expose a clear label and a reason in the UI and reports/exports, and the number of confirmed anomalies is zero.

Traceability: SC4, D2, M2; SC4, D4, M3; SC4, D6, M5

Verification: Test, Demonstration

Requirement ID: UR-05

Title: Historical navigation (timeline of runs)

Description: The user shall browse historical runs for a parcel via a date/timeline selector and open past results to review field evolution over time.

Rationale: Enables longitudinal interpretation and supports operational vineyard monitoring across successive acquisitions.

Dependencies: UI-04, API-04, FR-16

Acceptance Criteria: For a parcel with multiple runs, the UI lists or timelines available acquisitions (acquisition_date/scene_id) and allows opening a past run to display its outputs and exports consistently with UR-02 and UR-03.

Traceability: D4, M3

Verification: Demonstration, Inspection

External Interface Requirements - Structured**User Interface (UI)****Requirement ID: UI-01**

Title: Parcel definition and validation feedback

Description: The UI shall allow parcel creation by drawing or importing GeoJSON/Shapefile and shall provide immediate validation feedback against the AOI contract on invalid inputs.

Rationale: Enables correct AOI acquisition and prevents invalid runs.

Dependencies: FR-02, FR-03, DATA-01

Acceptance Criteria: Invalid geometry triggers explicit validation errors; valid parcels can be saved and re-opened across sessions.

Traceability: SC5, D4, M3

Verification: Test, Demonstration

Requirement ID: UI-02

Title: Run submission and run status (acquisition-anchored)

Description: The UI shall allow the user to submit a run for the latest available acquisition (and, if enabled, a selected past acquisition) and observe run status through terminal completion.

Rationale: Makes the on-demand workflow operable and inspectable while keeping the acquisition-anchored model explicit.

Dependencies: API-01, API-02, FR-04, FR-05

Acceptance Criteria: Status is displayed using: QUEUED, RUNNING, SUCCESS, FAILED, LOW_CONFIDENCE_DONE, and FAILED shows a structured failure stage/reason while LOW_CONFIDENCE_DONE shows a low-confidence label/reason.

Traceability: SC1, D3, M3; SC4, D4, M3; SC1, D6, M5

Verification: Demonstration, Inspection

Requirement ID: UI-03

Title: Map-first results and exports for a selected run

Description: The UI shall present outputs in a map-first layout and provide export access to PDF/CSV/GeoJSON for the currently selected run.

Rationale: Completes the “view, export” loop for operational use.

Dependencies: API-03, FR-12, FR-13

Acceptance Criteria: For any terminal run, the anomaly layers, macro report, and reliability indicator are visible, and PDF/CSV/GeoJSON exports are accessible for download.

Traceability: SC2, D2, M2; SC2, D5, M4; SC5, D4, M3; SC4, D4, M3

Verification: Demonstration, Inspection

Requirement ID: UI-04

Title: Historical navigation (date/timeline selector)

Description: The UI shall provide a date/timeline selector that lists available runs for the selected parcel and allows switching the displayed map layers to any past run without reloading the application context.

Rationale: Enables users to review field evolution across successive acquisitions and compare outcomes over time.

Dependencies: API-04, FR-16

Acceptance Criteria: For a parcel with multiple runs, the UI lists runs by acquisition_date/scene_id and selecting a past run updates the map layers and summary panels to that run’s outputs within the same map view.

Traceability: D4, M3

Verification: Demonstration, Inspection

Requirement ID: UI-05

Title: Reliability surfacing and LOW_CONFIDENCE distinction

Description: The UI shall surface observation reliability consistently (reliability indicator and/or reliability map) and visually distinguish LOW_CONFIDENCE_DONE runs from normal runs in both the run list and the map view.

Rationale: Prevents over-trusting low-quality EO observations and reinforces conservative interpretation.

Dependencies: API-03, FR-10, NFR-04

Acceptance Criteria: LOW_CONFIDENCE_DONE runs are clearly labeled in the run list and map view, show a non-empty reason/rationale, and the UI avoids presenting anomalies as confirmed under low confidence.

Traceability: SC4, D2, M2; SC4, D4, M3; D6, M5

Verification: Test, Demonstration

Requirement ID: API-01

Title: Create run (acquisition-anchored)

Description: The API shall accept a run request for a parcel and a target acquisition selector (default: latest available) and return a unique run_id.

Rationale: Establishes the on-demand execution entrypoint and traceable run identity in an acquisition-anchored model.

Dependencies: FR-04, FR-05

Acceptance Criteria: 100% of valid requests return a run_id and an initial persisted status QUEUED; the created run record includes acquisition selector fields (and, once resolved, acquisition_date and/or scene_id).

Traceability: SC1, D3, M3; SC5, D3, M3

Verification: Integration test

Requirement ID: API-02

Title: Get run status (with failure and low-confidence reasons)

Description: The API shall return the current run status and, if FAILED, a structured failure reason; if LOW_CONFIDENCE_DONE, a structured low-confidence rationale.

Rationale: Enables deterministic UI status reporting, debuggability, and conservative interpretation under poor EO conditions.

Dependencies: FR-05, FR-15, FR-10

Acceptance Criteria: Status reflects the persisted lifecycle state; FAILED includes stage and reason (and error_code when defined); LOW_CONFIDENCE_DONE includes reliability_factor, threshold, decision, and a non-empty reason.

Traceability: SC1, D3, M3; SC4, D3, M3; SC1, D6, M5

Verification: Integration test

Requirement ID: API-03

Title: Get run results, artifacts, and metadata

Description: The API shall return run outputs (map layer descriptors and artifact references), reliability indicator/surface references, and core run metadata for a given run_id.

Rationale: Allows serving results while preserving auditability and reproducibility via stable contracts.

Dependencies: FR-12, FR-13, FR-14

Acceptance Criteria: For terminal runs, the payload contains references to the three mandatory outputs and exports, a reliability summary, acquisition anchoring (acquisition_date and/or scene_id), and the core metadata keys required for traceability (versions, parameters, provenance, artifact keys).

Traceability: SC2, D3, M3; SC2, D5, M4; SC5, D4, M3

Verification: Integration test

Requirement ID: API-04

Title: List runs for parcel (history/timeline navigation)

Description: The API shall list available runs for a parcel, including acquisition_date and/or scene_id, terminal status, reliability summary, and the minimal references needed for UI timeline navigation.

Rationale: Enables historical browsing and map-first switching across past runs without requiring user-defined time windows.

Dependencies: FR-16, FR-14

Acceptance Criteria: For a parcel with N runs, the API returns an ordered list (by acquisition date) where each entry includes run_id, acquisition_date/scene_id, status, reliability_factor (or decision), and a pointer to retrieve results/artifacts via API-03.

Traceability: D3, M3; D4, M3

Verification: Integration test, Inspection

Data Requirements (DATA)

Requirement ID: DATA-01

Title: AOI contract

Description: AOI inputs shall conform to a defined GeoJSON schema and validation rules.

Rationale: Ensures interoperability and deterministic AOI handling.

Dependencies: FR-02

Acceptance Criteria: Invalid AOIs are rejected with explicit, structured error messages; valid AOIs are stored as canonical GeoJSON and produce a stable aoi_id and aoi_hash.

Traceability: SC5, D1, M1; SC5, D4, M3

Verification: Test, Inspection

Requirement ID: DATA-02

Title: Run bundle contract (acquisition-anchored)

Description: Each terminal run shall produce a self-contained run bundle containing the three mandatory outputs and exports, anchored to a specific acquisition (acquisition_date and/or scene_id), with stable naming/paths and artifact references for reproducibility.

Rationale: Makes SC2 verifiable and supports audit, rerun determinism, and historical navigation.

Dependencies: FR-12, FR-13, FR-14

Acceptance Criteria: For every terminal run, the run bundle contains all mandatory outputs and exports under stable logical keys, and run metadata includes acquisition anchoring fields and artifact references/paths/checksums consistent with the bundle layout. Bundle completeness is checked automatically (CI regression test).

Traceability: SC2, D2, M2; SC2, D5, M4

Verification: Test (CI), Inspection

Requirement ID: DATA-03

Title: Export validity and required attributes

Description: Exported GeoJSON shall be syntactically valid and include required attributes, including run_id, parcel_id (or aoi_id), acquisition_date and/or scene_id, status, reliability_factor, confidence_label (including LOW_CONFIDENCE when applicable), and the macro anomaly summary fields defined by the export contract.

Rationale: Guarantees GIS interoperability and testability of exports.

Dependencies: FR-13, FR-14

Acceptance Criteria: GeoJSON loads in a GIS viewer and passes schema/required-field checks; required attributes are present and consistent with run metadata and other exports (macro CSV, reliability JSON).

Traceability: SC2, D2, M2; SC2, D5, M4
Verification: Test, Demonstration, Inspection

System Functional Requirements (FR) - Structured

Requirement ID: FR-01

Title: Authentication

Description: The system shall authenticate users for UI/API access.

Rationale: Protects non-public resources and enables per-user parcel/run ownership.

Dependencies: -

Acceptance Criteria: Unauthorized requests are rejected (401/403).

Traceability: SC5, D3, M3; SC5, D4, M3

Verification: Test

Requirement ID: FR-02

Title: Parcel creation/import and validation

Description: The system shall support parcel creation (draw) and import (GeoJSON/Shapefile) with validation.

Rationale: Ensures AOI correctness and reduces run failures due to invalid inputs.

Dependencies: DATA-01

Acceptance Criteria: Invalid inputs fail with explicit reasons; valid parcels are persisted.

Traceability: SC5, D1, M1; SC5, D4, M3

Verification: Test, Demonstration

Requirement ID: FR-03

Title: Parcel persistence (multiple parcels)

Description: The system shall allow saving and retrieving multiple parcels per user across sessions.

Rationale: Enables iterative monitoring across multiple parcels and repeated use.

Dependencies: FR-01

Acceptance Criteria: A user can store and retrieve multiple parcels (≥ 3) and run analyses on each.

Traceability: SC5, D4, M3

Verification: Demonstration, Inspection

Requirement ID: FR-04

Title: Run creation (on-demand)

Description: The system shall create a run record and unique run_id upon user submission.

Rationale: Defines the unit of execution and traceability; enforces the on-demand model.

Dependencies: -

Acceptance Criteria: No run is executed without an explicit user request; each valid request creates a persisted run record with run_id and initial status QUEUED.

Traceability: SC1, D3, M3; SC1, D6, M5; SC5, D3, M3

Verification: Test, Inspection

Requirement ID: FR-05

Title: Run lifecycle management

Description: The system shall manage run statuses including QUEUED, RUNNING, SUCCESS, FAILED, and LOW_CONFIDENCE_DONE.

Rationale: Supports UI and monitoring consistency and deterministic terminal outcomes.

Dependencies: -

Acceptance Criteria: Status transitions are timestamped; FAILED includes stage and reason; LOW_CONFIDENCE_DONE includes a structured low-confidence rationale.

Traceability: SC1, D3, M3; SC4, D3, M3; SC1, D6, M5

Verification: Test, Inspection

Requirement ID: FR-06

Title: EO ingestion (acquisition-anchored)

Description: For each run, the system shall ingest required EO data and static layers for the AOI and the target acquisition (default: latest available), identified by acquisition_date and/or scene_id; optional minimal contextual acquisitions may be retrieved when required for stabilization/tracking, without being a user-defined time window.

Rationale: Provides the minimum signal needed for the MVP analytics kernel while keeping execution deterministic and auditable.

Dependencies: FR-04, DATA-01

Acceptance Criteria: Run metadata records dataset identifiers and acquisition anchoring (acquisition_date/scene_id) for the target acquisition, and records any additional contextual acquisitions if used.

Traceability: SC1, D1, M1; SC2, D1, M1

Verification: Inspection, Test

Requirement ID: FR-07

Title: Preprocessing and QA

Description: The system shall apply preprocessing and QA (masking, harmonization, alignment to the reference grid, missingness rules) prior to analysis.

Rationale: Prevents spurious anomalies due to invalid pixels or misalignment.

Dependencies: FR-06

Acceptance Criteria: QA outputs include valid-pixel and coverage statistics and are recorded in run metadata.

Traceability: SC1, D1, M1; SC4, D2, M2

Verification: Test, Inspection

Requirement ID: FR-08

Title: Feature computation (core indices)

Description: The system shall compute core indices/features required by the detector.

Rationale: Provides stable derived inputs to anomaly scoring.

Dependencies: FR-07

Acceptance Criteria: Feature artifacts exist, are aligned to the declared grid, and are referenced in run metadata.

Traceability: SC1, D1, M1; SC2, D2, M2

Verification: Inspection, Test

Requirement ID: FR-09

Title: Anomaly scoring (unsupervised-first)

Description: The system shall compute a pixel-level anomaly/deviance score map without

requiring ground-truth labels.

Rationale: Enables MVP operation without labeled datasets.

Dependencies: FR-08

Acceptance Criteria: Runs complete and produce anomaly outputs with no labels provided; reruns on the same target acquisition are deterministic with respect to acquisition anchoring and artifact keys.

Traceability: SC2, D2, M2

Verification: Demonstration, Test

Requirement ID: FR-10

Title: Coverage-reliability factor

Description: The system shall compute a coverage-reliability factor for every run.

Rationale: Quantifies observation adequacy and drives conservative behavior.

Dependencies: FR-07

Acceptance Criteria: Reliability fields (factor, decision, threshold, reason when low) are present in outputs/metadata for 100% of terminal runs.

Traceability: SC4, D2, M2; SC4, D6, M5

Verification: Test, Inspection

Requirement ID: FR-11

Title: Conservative confirmation policy

Description: The system shall confirm anomalies only when configured consistency/ensemble rules are satisfied and observation reliability is adequate; LOW_CONFIDENCE_DONE runs shall not present anomalies as confirmed.

Rationale: Controls false positives in unsupervised detection and prevents misleading interpretation under poor EO conditions.

Dependencies: NFR-03, NFR-04, FR-10

Acceptance Criteria: Policy test suite shows zero confirmation violations; confirmed anomalies count is zero under LOW_CONFIDENCE_DONE.

Traceability: SC3, D5, M4; SC3, D2, M5; SC4, D2, M5

Verification: Test, Inspection

Requirement ID: FR-12

Title: Mandatory outputs (SC2-aligned)

Description: For each terminal run, the system shall generate the following mandatory outputs: anomaly heatmap (spatial layer), macro report, and reliability factor (and/or reliability surface reference).

Rationale: Fixes an auditable minimal deliverable set for grading and user value.

Dependencies: -

Acceptance Criteria: Output completeness check passes for the demo campaign; artifacts include acquisition anchoring fields in metadata/exports.

Traceability: SC2, D2, M2; SC2, D5, M4; SC4, D2, M2

Verification: Test, Inspection

Requirement ID: FR-13

Title: Exports (PDF/CSV/GeoJSON)

Description: The system shall generate PDF/CSV/GeoJSON exports for each terminal run.

Rationale: Supports sharing, downstream analysis, and GIS interoperability.

Dependencies: DATA-03

Acceptance Criteria: Exports exist and pass validity checks; exports include required attributes including acquisition_date/scene_id and confidence labeling.

Traceability: SC2, D2, M2; SC2, D5, M4; SC4, D2, M2

Verification: Test, Demonstration

Requirement ID: FR-14

Title: Run metadata for reproducibility

Description: The system shall store run metadata sufficient for reproducibility and auditability, including parcel reference, acquisition_date/scene_id, quality/reliability summaries, pipeline/detector/policy versions, parameters, and artifact references inside the run bundle.

Rationale: Enables auditability and reproducible re-execution.

Dependencies: DATA-02, API-03

Acceptance Criteria: Given run_id, metadata is retrievable and sufficient to audit the run end-to-end (inputs, anchoring, parameters, versions, artifacts).

Traceability: SC2, D2, M2; SC2, D5, M4; SC1, D6, M5

Verification: Inspection, Test

Requirement ID: FR-15

Title: Structured errors

Description: FAILED runs shall include structured error information (stage, reason).

Rationale: Enables debugging, observability, and deterministic failure classification.

Dependencies: NFR-06, API-02

Acceptance Criteria: 100% of failures include stage and reason (and error_code if defined) in metadata and API responses.

Traceability: SC1, D3, M3; SC1, D6, M5

Verification: Test, Inspection

Requirement ID: FR-16

Title: Historical navigation support (run listing)

Description: The system shall support historical navigation by listing available runs for a parcel (ordered by acquisition date), including terminal status, reliability summary, and references required to retrieve outputs and exports for each run.

Rationale: Enables timeline-based review of field evolution and consistent UI navigation without user-defined time windows.

Dependencies: API-04, FR-14

Acceptance Criteria: For a parcel with multiple runs, the system returns an ordered run list where each entry includes run_id, acquisition_date/scene_id, status, and reliability summary, and each run can be opened via the results endpoint.

Traceability: D3, M3; D4, M3

Verification: Integration test, Demonstration

Requirement ID: FR-17

Title: Zone/cluster tracking representation (cross-run)

Description: The system shall produce and persist a tracking representation across successive runs that supports zone/cluster evolution summaries (e.g., persistence, emergence, merge/split) for historical interpretation in the UI.

Rationale: Supports a less noisy, agronomist-friendly view of anomalies and enables meaningful

historical interpretation beyond pixel-level volatility.

Dependencies: FR-09, FR-14, FR-16

Acceptance Criteria: When two or more runs exist for a parcel, the system produces a tracking artifact or summary fields that allow the UI to present stable zone-level evolution indicators across acquisitions.

Traceability: D2, M5; D4, M3

Verification: Inspection, Demonstration

4. Non-Functional Requirements (NFR)

4.1 NFR Summary

- **NFR-01:** On 20 valid runs, $\geq 95\%$ reach a terminal state and produce a complete run bundle.
- **NFR-02:** Every terminal run produces the run bundle with 3 mandatory outputs and exports, anchored to a specific acquisition (acquisition date and/or scene identifier).
- **NFR-03:** No confirmed anomaly violates the configured consistency/ensemble rules.
- **NFR-04:** Low reliability or insufficient observation adequacy triggers LOW_CONFIDENCE_DONE and prevents anomalies from being presented as confirmed.
- **NFR-05:** A guided user completes the workflow ≤ 30 minutes.
- **NFR-06:** Runs emit logs/metrics sufficient to compute success rate, failure-by-stage counts, and low-confidence incidence, attributable to the target acquisition and processing stage.

4.2 Structured NFRs

Requirement ID: NFR-01

Title: Run reliability (SC1)

Description: On a predefined test campaign of 20 valid run requests, $\geq 95\%$ reach a terminal state and produce a complete run bundle.

Rationale: Ensures the MVP is operationally reliable for evaluation and user trust.

Dependencies: FR-03, FR-04, FR-05, FR-12, FR-13, DATA-02

Acceptance Criteria: Test campaign logs show $\geq 19/20$ runs reaching a terminal state (SUCCESS, FAILED, or LOW_CONFIDENCE_DONE) and producing a run bundle suitable for inspection.

Traceability: SC1, D1, M1; SC1, D3, M3; SC1, D6, M5

Verification: Test, Inspection

Requirement ID: NFR-02

Title: Output completeness gate (SC2, acquisition-anchored)

Description: Every terminal run produces the run bundle with three mandatory outputs and exports, anchored to a specific acquisition (acquisition_date and/or scene_id).

Rationale: Makes completeness objectively verifiable and CI-enforceable under the acquisition-anchored execution model.

Dependencies: DATA-02, FR-12, FR-13, FR-14

Acceptance Criteria: CI regression test fails if any mandatory artifact/export is missing or if acquisition anchoring fields are absent from run metadata/exports for terminal runs.

Traceability: SC2, D2, M2; SC2, D5, M4

Verification: Test (CI), Inspection

Requirement ID: NFR-03

Title: Policy invariant (SC3)

Description: No confirmed anomaly violates the configured consistency/ensemble rules.

Rationale: Prevents misleading “confirmed” claims in an unsupervised setting.

Dependencies: FR-11

Acceptance Criteria: Automated policy tests report zero violations across the test campaign; confirmed anomalies are never present under LOW_CONFIDENCE_DONE.

Traceability: SC3, D5, M4; SC3, D2, M5; SC4, D2, M5

Verification: Automated test, Inspection

Requirement ID: NFR-04

Title: Low-confidence behavior (SC4)

Description: Low reliability or insufficient observation adequacy triggers LOW_CONFIDENCE_DONE and prevents anomalies from being presented as confirmed.

Rationale: Enforces conservative/no-decision behavior under poor EO adequacy.

Dependencies: FR-10, FR-11, UR-04, DATA-02

Acceptance Criteria: A forced low-adequacy test yields LOW_CONFIDENCE_DONE; confirmed anomalies count = 0; mandatory outputs/exports are still produced; low-confidence reason is present in reliability fields and exports.

Traceability: SC4, D2, M2; SC4, D4, M3; SC4, D6, M5

Verification: Test, Demonstration, Inspection

Requirement ID: NFR-05

Title: Usability baseline (SC5)

Description: A guided user completes the workflow within ≤ 30 minutes.

Rationale: Ensures the MVP is demonstrable without developer intervention.

Dependencies: UR-01, UI-01, UI-02, UI-03, UI-04, UI-05

Acceptance Criteria: Timed guided demo completes within ≤ 30 minutes end-to-end, including selecting latest run, opening results, browsing history, and downloading exports.

Traceability: SC5, D4, M3

Verification: Demonstration

Requirement ID: NFR-06

Title: Minimal observability (acquisition-attributed)

Description: Runs emit logs/metrics sufficient to compute success rate, failure-by-stage counts, and low-confidence incidence, attributable to the target acquisition and processing stage.

Rationale: Enables evidence-based evaluation, monitoring, and debugging.

Dependencies: FR-15, FR-14, FR-05

Acceptance Criteria: For the test phase, logs/metrics allow counting terminal states, grouping failures by stage, and reporting LOW_CONFIDENCE_DONE rates with acquisition_date/scene_id attribution.

Traceability: SC1, D6, M5; SC4, D6, M5

Verification: Inspection, Test

5. Project Architecture

This section explains how the MVP works end-to-end and where verification evidence is produced. The architecture is explicitly run-based and SC1-SC4 are verified by inspecting run outcomes, artifacts, and logs.

5.1 Component view

SmartHarvest MVP is decomposed into the following logical components:

1. **Web Application (UI):** interactive map, parcel management, on-demand run submission (default: latest acquisition), run status and historical timeline, results visualization (anomaly layers, feature layers, reliability surfaces/indicators), and export downloads.
2. **API Service:** authentication, parcel endpoints, run lifecycle endpoints (create/status/results), run listing for historical navigation, and serving of results/artifacts/exports.
3. **Orchestrator:** asynchronous execution of user-triggered runs anchored to a target acquisition; optional scheduled jobs limited to data refresh checks and housekeeping when enabled (does not replace the on-demand model).
4. **Data Access Layer:** retrieval of EO observations and ancillary layers via a managed EO platform, including identification of the latest available acquisition and retrieval by acquisition date/scene identifier.
5. **Preprocessing and QA Module:** masking/filtering, CRS/resampling alignment, acquisition-level adequacy and quality statistics, feature harmonization across acquisitions, and (when enabled) minimal-context stabilization rules required for tracking (without user-defined time windows or long-window regularization).
6. **ML Kernel:** unsupervised spatial structure discovery (zones/clusters), computation of per-pixel and zone-level anomaly outputs, macro summary metrics, coverage-reliability factor, and tracking summaries across successive acquisitions where applicable
7. **Storage Layer:** persistence of parcels, run registry, run metadata, artifacts/outputs, exports, and the tracking representation used for historical interpretation.
8. **Monitoring and Logging:** structured run events and KPI computation (run health, data-quality signals, low-confidence incidence, and policy compliance indicators).

5.2 Execution model and serving

- **Execution is on-demand:**

A run is created only by explicit user request, then transitions through a stored lifecycle until it reaches a terminal state (API-02, FR-05).

The ML kernel executes as a batch job per run, anchored to a target acquisition (default: latest available), and is orchestrated asynchronously to keep the UI responsive while providing observable status and structured failure/low-confidence reasons (FR-08).

- **Serving is contract-driven:**

The API exposes stable endpoints for run creation, status, run listing for historical navigation, and results/artifacts retrieval (IF-API-01/02/03, API-04). The UI consumes the same contracts for visualization, timeline-based browsing, and downloads (IF-UI-01/02).

A single-demo command execution shall reproduce the end-to-end workflow and generate a complete run bundle suitable for inspection against SC2 and the declared artifact contract.

5.3 Monitoring

Monitoring is a core part of the MVP detection strategy: since the kernel is unsupervised-first and scene-anchored, the project is evaluated through operational behavior, observation adequacy, and contract/policy compliance, not label-based accuracy. The system therefore records run events and a set of run-level metrics to verify Success Criteria 1-4.

Minimum monitoring signals include:

1. **Operational monitoring:** run success rate and terminal-state distribution; latency/runtimes per acquisition and pipeline stage (including p95 where applicable); failures by stage; queue/backlog indicators.
2. **Data-quality monitoring:** LOW_CONFIDENCE rate trend per parcel and over time; valid-pixel ratio and missingness trends for each acquisition; cloud-masked ratio and coverage-reliability distributions; incidence of “insufficient observation” causes by type.
3. **Analytics health proxies:** stability of macro summary statistics for high-confidence runs across successive acquisitions; stability of detected zone/cluster structure (e.g., excessive churn rate, frequent merge/split spikes) as a proxy for robustness; frequency of no-decision outcomes.

Monitoring summaries constitute objective evidence that the system behaves conservatively under poor EO conditions (SC4) and maintains expected reliability and completeness properties over the test campaign (SC1-SC2).

5.4 Architecture constraints and assumptions

- **EO platform assumption:** MVP EO access and compute are assumed to rely on a managed platform (e.g., Google Earth Engine).
- **Stable contracts:** UI, API, and run bundle contracts shall remain compatible across kernel iterations, including acquisition-anchored run identifiers/metadata, layer descriptors for visualization, and export schemas used by the UI and downstream tooling.
- **Security baseline:** all non-public resources shall require authentication and shall use TLS for data in transit (FR-01).

6. Risk Analysis

This section identifies the main risks that can affect trustworthiness, reproducibility, usability, and assessability of the MVP of SmartHarvest. Each risk is matched to a mitigation strategy and observable indicators to enable detection and corrective action.

6.1 Risk rating and prioritization

Probability (P) and Impact (I) are rated on a 1 to 5 ordinal scale for the MVP development and risk evaluation.

P: 1 Rare, 2 Unlikely, 3 Possible, 4 Likely, 5 Almost certain.

I: 1 Negligible, 2 Minor, 3 Moderate, 4 Major, 5 Critical.

Risk Exposure (RE) is computed as $[RE = P \times I]$ and used to prioritize mitigation and verification. Items with $RE \geq 15$ are treated as high priority; 10-14 medium; ≤ 9 low.

Risk	Consequence	Prob	Impact	RE	Mitigation technique	Detection
R-01: Cloud cover / missing EO observations (acquisition-level)	Unstable outputs or misleading interpretation of field state	5	4	20	Enforce acquisition-level observation adequacy rules and a coverage-reliability factor; downgrade to LOW_CONFIDENCE_DO NOT / no-decision behavior when valid-pixel coverage or quality is insufficient	Increasing LOW_CONFIDENCE rate; valid-pixel ratio trend; cloud-masked ratio; missingness/coverage distributions per acquisition.
R-02: EO provider quota/outage (platform limits)	Ingestion failures and broken run execution	3	4	12	Retry policy with backoff; stage-level failure recording; cache static layers; allow re-run once the acquisition is available; degrade gracefully when provider errors occur.	Spike in ingestion failures; error logs concentrated in ingestion stage; quota-limit events.
R-03: Misalignment / resampling across sources	False spatial anomalies and inconsistent maps	3	4	12	Strict CRS and resampling policy; declared reference grid; alignment checks on every raster artifact; automated tests on geometries/rasters.	CI failures on alignment checks; spatial discontinuity artifacts; sudden striping/edge effects correlated with a specific layer.
R-04: Kernel/Trackin	Inconsistent anomaly	3	4	12	Anchor each run to acquisition date/scene_id	Rerun-consistency checks on same

g instability across acquisitions (or non-determinism on same acquisition)	outputs, excessive zone churn, hard-to-interpret history				and enforce deterministic preprocessing; store full provenance and versions; apply stable cluster tracking association rules.	acquisition; drift in macro summaries without data change; cluster churn spikes (merge/split frequency).
R-05: False positives overload (especially under noisy observations)	User distrust and abandonment	3	5	15	Conservative confirmation policy; condition “confirmed” on reliability/adequacy; expose reliability clearly in UI/exports; avoid causal claims.	Confirmed anomalies always pass policy tests; confirmed-anomaly rate vs reliability.
R-06: Runtime too high for larger AOIs	Demo failure or unusable workflow	3	4	12	Define AOI size constraints for evaluation; performance tests; cache static layers; avoid unnecessary recomputation; process per acquisition; asynchronous orchestration to keep UI responsive.	p95 runtime exceeds target; latency by stage; queue/backlog growth; orchestrator timeouts.
R-07: Artifact incompleteness / partial run bundles	Expected output violation and non-assessable runs	2	5	10	CI regression checks; atomic write strategy for run bundle; artifact presence checks before marking terminal.	CI regression failures; export schema validation fails; invariant violations (terminal state but missing artifacts).
R-08: Non-repeatable deployment / environment drift	System cannot be run or evaluated reliably	3	5	15	Containerization; pinned dependencies; environment template; single demo command; README with reproducible setup; versioned contracts for API/UI/run bundle.	Clean-machine run fails; Docker build fails; demo command doesn’t produce a complete run bundle; contract mismatch errors at runtime.

EO data availability and quality is the dominant risk: cloud-driven gaps can make outputs unstable or misleading, so the MVP enforces acquisition-level observation adequacy rules and uses a coverage-reliability factor to trigger LOW_CONFIDENCE_DONE or a no-decision behavior instead of presenting confirmed anomalies when reliability is insufficient.

Continuity risks (EO platform quota/outage, misalignment/resampling across sources) are handled via retry policies with backoff, stage-level failure recording, caching of static layers where applicable, and strict grid-alignment checks against the declared reference grid.

Because the kernel is unsupervised-first and scene-anchored, trust risks (non-determinism on the same acquisition, tracking instability across acquisitions, and false positives) are mitigated through

deterministic preprocessing and fixed kernel parameters, conservative confirmation conditioned on observation reliability, and reproducibility controls that record provenance, versions, and run configuration inside the run bundle.

Finally, evaluation risks (missing artifacts, non-repeatable deployment) are mitigated through CI regression gates (artifact presence and export schema validation), atomic artifact writing, containerization with pinned dependencies, and a single demo command that consistently produces a complete run bundle for verification.

Appendix A: Traceability and Verification Summary

Appendix A provides minimal but explicit traceability links between Success Criteria (SC), requirements, WBS and verification evidence.

1. Success Criteria to Requirements

Success Criteria	Primary requirements
SC1 Run reliability ($\geq 95\%$ terminal states)	FR-04/05/15, API-01/02, NFR-01/06, FR-14
SC2 Output + export completeness (per terminal run)	FR-12/13/14, UI-03, API-03, DATA-02/03, NFR-02
SC3 Conservative confirmation policy (no violations)	FR-11, NFR-03, NFR-04, UR-04
SC4 Missing data robustness (LOW_CONFIDENCE_DONE and no confirmations)	FR-05/10/11/12/13, UR-04, API-02, NFR-04
SC5 Usability (≤ 30 min guided session)	UR-01/02/03/04/05, UI-01/02/03/04/05, API-01/02/03/04, NFR-05, FR-01

2. Use Case to Requirements

Use Case ID	Use Case (SSD scope)	Covered by:
UC-01	Define / import parcel (AOI) and persist it	UR-01, UI-01, DATA-01, FR-02, FR-03
UC-02	Submit on-demand run for a parcel (acquisition-anchored; default: latest available)	UR-01, UI-02, API-01, FR-04, FR-05
UC-03	Monitor run lifecycle states and errors	UR-01, UI-02, API-02, FR-05, FR-15, NFR-06
UC-04	Visualize run outputs (map-first) and switch layers for the selected run	UR-02, UI-03, API-03, FR-12, FR-10
UC-05	Download exports (PDF/CSV/GeoJSON)	UR-03, UI-03, API-03, FR-13, DATA-03
UC-06 (optional)	Conservative confirmation behavior and low-confidence handling	UR-04, FR-10, FR-11, NFR-03, NFR-04
UC-07	Browse historical runs for a parcel (timeline/list) and open past results	UR-05, UI-04, API-04, FR-16

3. Deliverables to Requirements and WBS Anchors

Deliverable	Requirement anchors	WBS anchor
D1 - Reproducible Data Backbone	DATA-01, FR-06, FR-07, FR-08, FR-14	2.2, 2.3, 2.4, 2.5
D2 - ML Kernel Baseline and Outputs	FR-09, FR-10, FR-11, FR-12, FR-13	3.1, 3.2, 3.3, 3.4, 3.5, 3.6
D3 - Minimal Serving Layer (API)	API-01, API-02, API-03, API-04, FR-04, FR-05, FR-14, FR-15, FR-16	4.1, 4.2, 4.3
D4 - Web App	UR-01/02/03/04/05, UI-01/02/03/04/05, FR-02, FR-03	5.1, 5.2, 5.3, 5.4, 5.5
D5 - CI/CD and Packaging	NFR-02, NFR-03, NFR-04, DATA-02, DATA-03	6.1, 6.2, 6.3, 6.4
D6 - Monitoring Baseline and Auditability	NFR-06, FR-05, FR-14, FR-15	6.5, 6.4

Appendix B: Run Metadata Contract

This appendix defines the minimum run metadata fields that must be persisted for traceability, auditability, and reproducibility, consistent with the on-demand run model.

A run shall end in exactly one terminal state:

- **SUCCESS**: run completed and produced all mandatory outputs and exports.
- **LOW_CONFIDENCE_DONE**: run completed and produced all mandatory outputs and exports, but reliability is below threshold and **no anomalies may be confirmed**.
- **FAILED**: run terminated due to an error, failure stage and reason are recorded.

Non-terminal states:

- **QUEUED**: run accepted and persisted, awaiting execution.
- **RUNNING**: run execution started.

Section	Field	Type / Format	Notes / Example
Run identity & traceability	run_id	string	Unique run identifier
	created_at	ISO 8601 datetime	e.g., 2026-01-26T19:12:33Z
	status	enum	QUEUED RUNNING SUCCESS LOW_CONFIDENCE_DONE FAILED

Data provenance	data_sources	array	One entry per dataset/source used
	data_sources[].source_name	string	e.g., Sentinel-2 L2A
	data_sources[].provider	string	e.g., Copernicus/ESA
	data_sources[].image_count	integer	Number of EO observations actually used for the run (typically 1 for scene-anchored runs; >1 only if minimal deterministic context is used)
	data_sources[].acquisition_date	ISO 8601 date/datetime	Target acquisition date used by the run (preferred for scene-anchored MVP)
	data_sources[].scene_id	string	Provider/product identifier for the target acquisition (if available)
	data_sources[].acquisition_range.start	ISO 8601 date/datetime	Start of the effective acquisition range used (if a minimal context range is used internally)

	data_sources[].acquisition_range.end	ISO 8601 date/datetime	End of the effective acquisition range used (if a minimal context range is used internally)
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Quality and Reliability	quality	object	Run-level QA summary for the target acquisition (and any minimal deterministic context, if used)
	quality.acquisition_count	integer	Count of EO observations actually used by the run (typically 1 for scene-anchored runs; >1 only if minimal context is used)
	quality.valid_pixel_ratio	float [0,1]	Fraction of valid pixels after masking for the target acquisition
	quality.cloud_masked_ratio	float [0,1]	Fraction of pixels masked as cloud/shadow (only when cloud masking applies)
	reliability	object	Reliability decision + explanation
	reliability.reliability_factor	float [0,1]	Scalar adequacy indicator derived from quality/coverage signals
	reliability.threshold	number	Threshold used for LOW/HIGH decision
	reliability.decision	enum	HIGH_CONFIDENCE LOW_CONFIDENCE
	reliability.reason	string	Non-empty when LOW_CONFIDENCE (structured explanation suitable for UI/reporting)

Algorithm and Policy Versioning	feature_set_id	string	Identifier of the feature set/config used for this run (e.g., features_v1_s2_core)
	kernel_version	string	Version of the kernel (e.g., semantic version, git tag, or commit SHA)
	policy_version	string	Version of the confirmation + low-confidence policy applied (e.g., policy_v1_conservative)
	parameters	object	Key thresholds and detector/policy parameters used for reproducibility (e.g., reliability_threshold, clustering params, outlier cutoffs)

Artifacts and exports registry	artifacts	object	Logical keys → storage refs/checksums/paths
	artifacts.anomaly_heatmap	ref/path/checksum	Heatmap artifact

	artifacts.macro_report	ref/path/checksum	Macro report artifact
	artifacts.reliability_report	ref/path/checksum	Reliability report artifact
	artifacts.export_pdf	ref/path/checksum	PDF export
	artifacts.export_csv	ref/path/checksum	CSV export
	artifacts.export_geojson	ref/path/checksum	GeoJSON export

Failure observability (FAILED only)	failure	object	Present only if status = FAILED
	failure.stage	enum	INGEST PREPROCESS FEATURES DETECT RELIABILITY EXPORT SERVE
	failure.reason	string	Human-readable cause
	failure.error_code	string	Stable code if you define one
	failure.logged_at	ISO 8601 datetime	Timestamp of failure record

Appendix C: Data Contracts and Output Specifications

This appendix defines the minimum data contracts required to make the MVP interoperable, testable, and exportable. These contracts are intended to reduce ambiguity for verification.

1. AOI Contract

Accepted input

- AOI shall be provided as GeoJSON Polygon or MultiPolygon.
- AOI may be imported or created by drawing; both shall be stored as canonical GeoJSON.

Validation rules

- Geometry shall be valid (closed rings, non-self-intersecting).
- AOI shall be non-empty and have positive area.
- AOI shall be associated with an `aoi_id` and produce a stable `aoi_hash` (stable hash of canonical AOI geometry).
- CRS assumption shall be explicit (default EPSG:4326). If non-default CRS is accepted, the system shall record it and reproject deterministically.

Error reporting

Invalid AOIs shall be rejected with a structured error.

2. Mandatory run products

Each terminal run with status SUCCESS or LOW_CONFIDENCE_DONE shall produce exactly three mandatory outputs. All products shall be referenced in the run metadata using stable logical keys and shall be anchored to a specific EO acquisition (acquisition date and/or scene identifier).

1) Anomaly heatmap:

Logical meaning: per-cell anomaly score (continuous deviance score) for the target acquisition.

Physical form: GeoTIFF/COG aligned to the declared reference grid (Sentinel-2 10 m).

Minimum metadata: CRS, resolution, no-data policy, and declared score semantics (range and meaning), plus acquisition anchoring (acquisition_date and/or scene_id).

2) Macro report:

Logical meaning: parcel-level summary including percentage of anomalous pixels and required context fields for the target acquisition.

Physical form: CSV.

Minimum fields (CSV):

- run_id, aoi_id, acquisition_date (and/or scene_id)
- anomalous_pixel_pct (or ratio, but consistent across artifacts)
- valid_pixel_pct (or ratio, but consistent across artifacts)
- reliability_factor
- confidence_label (HIGH|LOW)
- status (SUCCESS|LOW_CONFIDENCE_DONE)

(Optional but recommended fields for auditability: pipeline_version, aoi_hash, no_data_policy_id)

3) Coverage-reliability report

Logical meaning: run-level reliability factor with explicit rationale and decision.

Physical form: JSON.

Minimum fields:

- run_id, acquisition_date (and/or scene_id)
- reliability_factor, threshold, decision, reason
- valid_pixel_ratio (or pct)
- cloud_masked_ratio (or pct, if available)
 - observation_count_used (number of EO observations actually used by the run; typically 1 for scene-anchored runs, >1 only if minimal context is used deterministically)

The system may optionally persist additional visualization artifacts (not required for terminal completeness), such as a zone-level/denoised anomaly layer and per-feature heatmaps. If produced, they shall be referenced in run metadata under separate, stable logical keys and must not replace the three mandatory outputs above.

3. Export formats

For each terminal run with status SUCCESS or LOW_CONFIDENCE_DONE, the system shall generate:

- **PDF report:** human-readable summary including map snapshot, macro report highlights, and an explicit reliability/low-confidence rationale when applicable.
 - **CSV export:** machine-readable run summary, at minimum the macro report fields.
 - **GeoJSON export:** spatial interoperability export. (Minimum content: AOI geometry and key run attributes (run_id, aoi_id, acquisition_date and/or scene_id, status, reliability_factor, confidence_label, and macro summary fields). It shall be valid GeoJSON, include the required attributes documented in this contract, and shall load in a GIS viewer.)
-

Appendix D: Glossary and Acronyms

- **AOI (Area of Interest):** The parcel boundary used for clipping EO data and producing run outputs (GeoJSON polygon).
- **Artifact:** Any persisted file produced by a run (inputs, intermediates, outputs, exports).
- **Coverage-reliability factor:** Scalar indicator (0-1) summarizing observation adequacy and data quality for the target acquisition.
- **CRS:** Coordinate Reference System used to locate spatial data (e.g., EPSG:4326).
- **Deviance score:** Continuous anomaly score used to color the anomaly heatmap (the higher, the more anomalous).
- **EO:** Earth Observation data (satellite observations).
- **GeoJSON:** JSON-based standard for representing geographic features.
- **Ingestion:** Retrieval of EO observations and ancillary datasets for a run, anchored to the target acquisition (acquisition date and/or scene identifier).
- **Low-confidence / no-decision:** Conservative behavior where the system produces outputs but avoids presenting anomalies as confirmed due to insufficient observation adequacy or poor EO quality for the target acquisition.
- **Macro report:** Parcel-level summary (e.g., % anomalous pixels, valid-pixel ratio, reliability factor) produced for each run.
- **On-demand execution:** Runs are created only when the user requests them (no mandatory schedule).
- **Preprocessing / QA:** Masking, harmonization, alignment to the reference grid, missingness rules, and quality checks applied before feature computation.
- **Run:** A single execution instance for a given AOI and target acquisition (default: latest available), producing a self-contained bundle of artifacts and metadata; the run is identified and reproducible via run_id and acquisition anchoring (acquisition date and/or scene identifier).

Appendix E: Process Flow Diagrams

The diagrams below define the operational workflows for SmartHarvest MVP:

- 1) The user interaction flow (parcel → run submission → results → exports)
- 2) The run execution lifecycle (QUEUED → RUNNING → terminal states), including status transitions and the evidence artifacts used for verification.

